

Original article

Efficiency of bidirectional EV charging: Key insights from implementing ISO 15118-20 CCS2 and CHAdeMO

Kristian Sevdari ^{a,*}, Kristoffer Laust Pedersen ^a, Gabriel Fabbri ^a, Rasmus Meier Knudsen ^{a,b}, Abdesslam Alami ^c, Jan Engelhardt ^a, Mattia Marinelli ^a

^a Department of Wind and Energy Systems, Technical University of Denmark, Frederiksborgvej 399, Building 330, 4000 Roskilde, Denmark

^b Monta, Strandboulevarden 122, 5, 2100 Copenhagen, Denmark

^c Watt&Well, 129, avenue de Paris, 91300 Massy, France

ARTICLE INFO

Keywords:

Bidirectional charging
Efficiency
Electric vehicles
Smart charging
Storage
V2G

ABSTRACT

Charging infrastructure serves as the interface between the electric mobility and power sectors. Here, the electrification of both sectors is of paramount importance, and the possibility of storing energy in the vehicle battery is crucial to coupling the synergy of such sectors. This paper presents a novel bidirectional DC charger equipped with CHAdeMO and CCS2 plugs, demonstrating successful integration and bidirectional power flow using the ISO 15118-20 standard. The charger features two 11 kW bidirectional modular power supplies and supports electric vehicle architectures of 400 V and 800 V. We tested the charger with two emulators and a commercial vehicle (Kia EV9), achieving successful integration via ISO 15118-20. However, complete transport layer security integration between the charger and vehicle remains a challenge. Moreover, the efficiency and power quality of the charger were characterized for the complete operating area. The results highlight that while the charger achieves high charging efficiency 80% (2 A) to 97% (32 A) for both CCS2 and CHAdeMO protocols, the discharging efficiency is notably lower, ranging from 60% (2 A) to 93% (32 A). The study demonstrates that this new generation of bidirectional chargers promises higher efficiency performance 7%–15% improvement compared to the previous CHAdeMO generation, enhancing the applicability of EVs as flexible energy storage units for behind or in-front-of-the-meter applications.

Introduction

Enhancing market penetration of EVs is crucial to achieving a sustainable energy transition within the transportation sector [1]. Beyond reducing the cost of EVs, identifying additional value streams can enhance their economic efficiency throughout their life cycle and incentivize broader adoption [2,3]. Although EVs are primarily purchased for mobility, they spend approximately 95% of their useful life parked and unused [4,5]. Leveraging this scenario, EVs can create a synergistic relationship between the transportation and power sectors by serving as energy storage units. Different investigations have concluded that vehicle-to-grid (V2G) greatly benefits the renewable energy integration (extra 40% PV integration) [6], alleviating congestion management [7], empowering end-users (prosumers) and improving power system resilience. For example, authors of [6] have underscored the great economic benefits for European users up to 780 Euro/year and save 22 billion Euro a year by 2040.

Bidirectional charging, which allows EVs to both draw and supply power, is not a new concept for EVs, as the Japanese CHAdeMO charging protocol allows such operation [1,8]. In addition, research projects leading to commercial applications [9] have been finalized in countries such as Japan, Denmark [10], the Netherlands [11], the United Kingdom [12], Australia, or the United States [13]. However, incentivized by the European Commission for a faster charging speed, the CHAdeMO protocol has been opted for combined charging system (CCS2) protocol. Thus, the penetration of such technology (CHAdeMO) in the market has been hampered. In the cases where CHAdeMO is running, concerns have been raised regarding vehicle battery degradation [14,15], user acceptance [16], regulation [17], technology efficiency [18], and business model [19]. These challenges persist even for the ISO 15118-20 standard that allows bidirectional charging for AC (e.g. Type 2) and DC (e.g., CCS2) plugs. In addition, the lack of commercially available vehicles able to do bidirectional charging with ISO 15118-20 has

* Corresponding author.

E-mail addresses: krisse@dtu.dk (K. Sevdari), klapeb@dtu.dk (K.L. Pedersen), s164551@dtu.dk (G. Fabbri), rakn@monta.com (R.M. Knudsen), abdesslam.alami@wattandwell.com (A. Alami), janen@dtu.dk (J. Engelhardt), matm@dtu.dk (M. Marinelli).

<https://doi.org/10.1016/j.seta.2025.104654>

Received 8 May 2025; Received in revised form 18 October 2025; Accepted 19 October 2025

Available online 30 October 2025

2213-1388/© 2025 Published by Elsevier Ltd.

limited the technology development. Also, the cost of BPT chargers still remains a barrier for customers.

ISO 15118

In addition to the aforementioned issues, the ISO 15118 standard has faced challenges due to its increased complexity. One can connect such difficulties with the fact that CCS2 was designed with fast charging as the main focus, and bidirectional charging is something added retroactively. Although CHAdeMO was primarily designed to enable bidirectional power transfer. Part 20 (15118-20) of ISO 15118, encompassing the 2nd Generation Network Layer and Application Layer Requirements, serves as a comprehensive advance over ISO 15118-2 [20]. It builds on Part 2, augmenting it with enhanced functionalities including more robust security protocols, wireless power transfer, and, critically, bidirectional power transfer. EVs and EVSEs adhering to ISO 15118-20 initiate baseline communications identically to their predecessors under Part 2. However, subsequent higher-level communications differ slightly due to newly integrated functionalities, making the two protocols incompatible [21]. Table A.1 describes the power transfer modes available with the ISO 15118-20 standard. In this paper we test the bidirectional power transfer according to service names “DC”, “DC_ACDP”, “DC_BPT”, and “DC_ACDP_BPT”. Another aspect of the challenge in the implementation of ISO 15118-20 is the security aspect, as known as transport layer security (TLS). To summarize, the challenges with ISO 15118-20 are the interoperability of communication and power flow and the security aspects.

Contributions

To the best of our knowledge, this is the first time an independent team has achieved full bidirectional compliance using ISO 15118-20 with an emulator and commercial vehicle (Kia EV9) and published the data. However, we are aware that there are ongoing collaborations behind closed doors between automakers such as Hyundai, Ford, General Electric, Volkswagen, Polestar, and charger manufacturers such as Wallbox, Ambibox/Intercontrol, and Easee. The difference is that automakers have demonstrated bidirectional charging with a customized implementation of ISO 15118-2, which hinders the interoperability between different automakers and charger manufacturers. As such, the first contribution of this paper is the demonstration of the Watt&Well bidirectional charger [22] from Technical University of Denmark (DTU) researchers against different units such as the Keysight Charging Discovery System emulator, the Trialog communication emulator, and different EV models.

The second contribution of the paper is the development of comprehensive efficiency maps of the bidirectional charger, covering a wide range of operating conditions. That includes a comparison of this new generation of charger with both the CHAdeMO and CCS2 (ISO 15118-20) plugs. As highlighted in [18,23–25], the efficiency of the bidirectional charger is of paramount importance for future adoption, as it affects business models and the feasibility against stationary energy storage. Previous work had overestimated the efficiency of on-board charger (OBC) and bidirectional chargers and their technological development [26,27]. However, recent thorough investigations have shown lower charging efficiencies and raised questions about smart charging use cases [28,29]. A review of the literature shows that the efficiency of bidirectional chargers can range between 10%–90%, depending on the operating setpoint [18,19,25,30–32]. Although an overview of bidirectional chargers is available in [25], this paper elevates the state-of-the-art to new technological improvements, such as a full-range test using 400 and 800 V architectures with CHAdeMO and CCS2. In addition, this paper contributes by utilizing state-of-the-art test equipment such as emulators and commercial vehicles. A comparison with previous research is shown in Table 1. From the table can be observed that our study provides data for the first time for

the ISO 15118-20 standard (CCS2 plug) and does so by testing the full range with commercial vehicles and test emulators. Consequently, different from the existing investigations, this paper presents data not just for the 400 V architecture but also for the 800 V.

Finally, the third contribution of this paper is to investigate the power quality behavior of the bidirectional charger under test. The power quality parameters, such as reactive power and power factor, are crucial to the operational security of the power grid. Previous studies have shown that these parameters often fall outside the acceptable range for commercial onboard chargers [28,34]. The remainder of the paper is structured as follows. “Methodology” describes the research methodology and the research infrastructure used to conduct the investigation campaign. In addition, “Investigation campaign results” presents the results from different test units. Finally, “Conclusions” concludes the article with the main findings.

Methodology

Technology and testing setup

The bidirectional charger in question is an off-board DC charger. That means the conversion from AC to DC occurs inside the bidirectional charger and the connection with EV is a DC connection. Therefore, to measure the efficiency of the bidirectional charger, is important to have measurements before (AC side) and after (DC side) power conversion. Fig. (1a) provides a detailed illustration of the test setup used in every case. We use PEL 104 to measure current and voltage from the AC side and L452 together with Hioki for the DC side. PEL 106 measures the auxiliary consumption of the bidirectional charge. The auxiliary consumption is considered in the efficiency calculations in “Investigation campaign results”. A Raspberry Pi synchronizes measurements across all devices with a digital voltage signal. The oscilloscope and Hioki datalogger are used for backup measurements and to quantify the response delays of the charger. Subsequently, the measurement devices and the uncertainty of the measurements are highlighted in Table A.2. Fig. 1(b) provides information on the test setups used in the testing campaign. The testing of the CHAdeMO and CCS2 plugs was carried out at two facilities: the DTU EV-laboratory in Denmark and ElaadNL in the Netherlands. ElaadNL and DTU e-mobility laboratories offer controlled 20–23 degrees Celsius conditions. Fig. 1(a) shows the experimental setups at both locations. ElaadNL provides the possibility to test bidirectional chargers using the Keysight charging discovery system (CDS) setup, which can emulate DC power up to 180 kW, with a voltage of up to 1000 V and a current of up to 300 A. The bidirectional charger itself has a rated power of 22 kW and a maximum voltage of 1000 V. ISO 15118-20 communication is performed via the Verisco/Keysight SL1550 A communication box to the CDS.

Fig. A.1 presents the block diagram of the bidirectional charger developed by Watt&Well. The bidirectional modular power supply (BMPU) can accommodate DC power of up to 11 kW, and a maximum voltage of 500 V. The Watt&Well charger under test has two BMPUs, which, when combined, can handle up to 22 kW at 1000 V on the DC side. This unique capability allows the Watt&Well bidirectional charger to support both 400 V and 800 V EV architectures [35]. Table A.3 provides a detailed explanation of the Watt&Well charger capabilities.

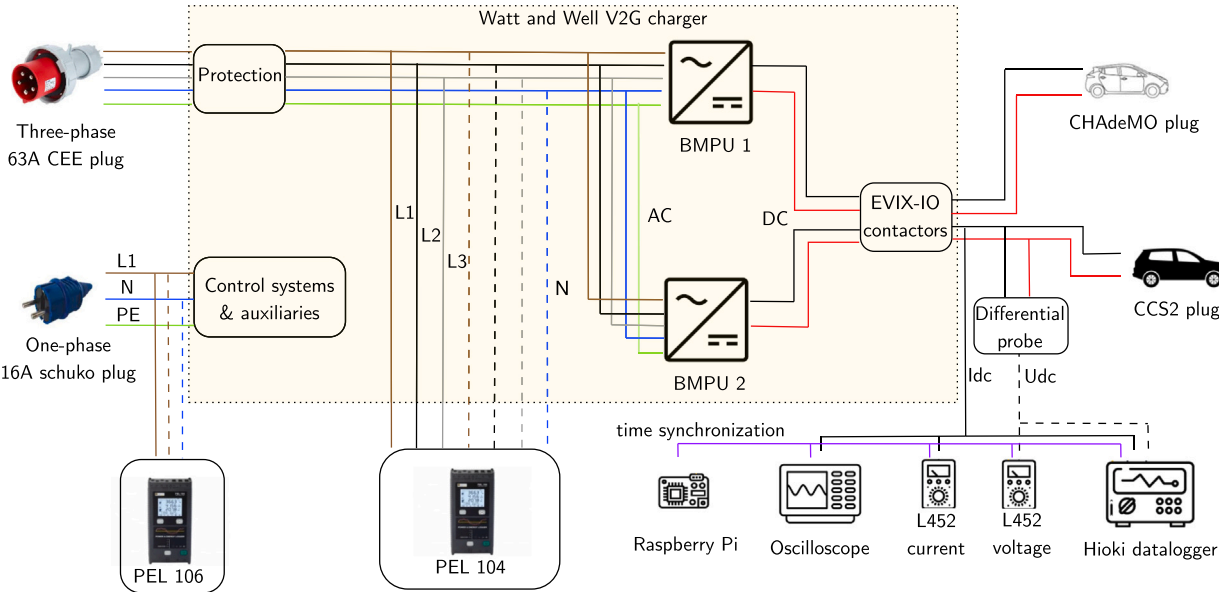
Performance indicators and test sequence

The test sequence starts from 0 kW and follows a linear path towards maximum charging power and afterwards to maximum discharging power. This sequence was intended to sweep the operating range of a given configuration of the vehicle and BMPU power converters. Such a pattern provides data for the determination of (ii) set-point linearity, (vi) accuracy, and (vii) precision as well as the additional metrics efficiency, reactive power and/or power factor, individual or total harmonic distortion.

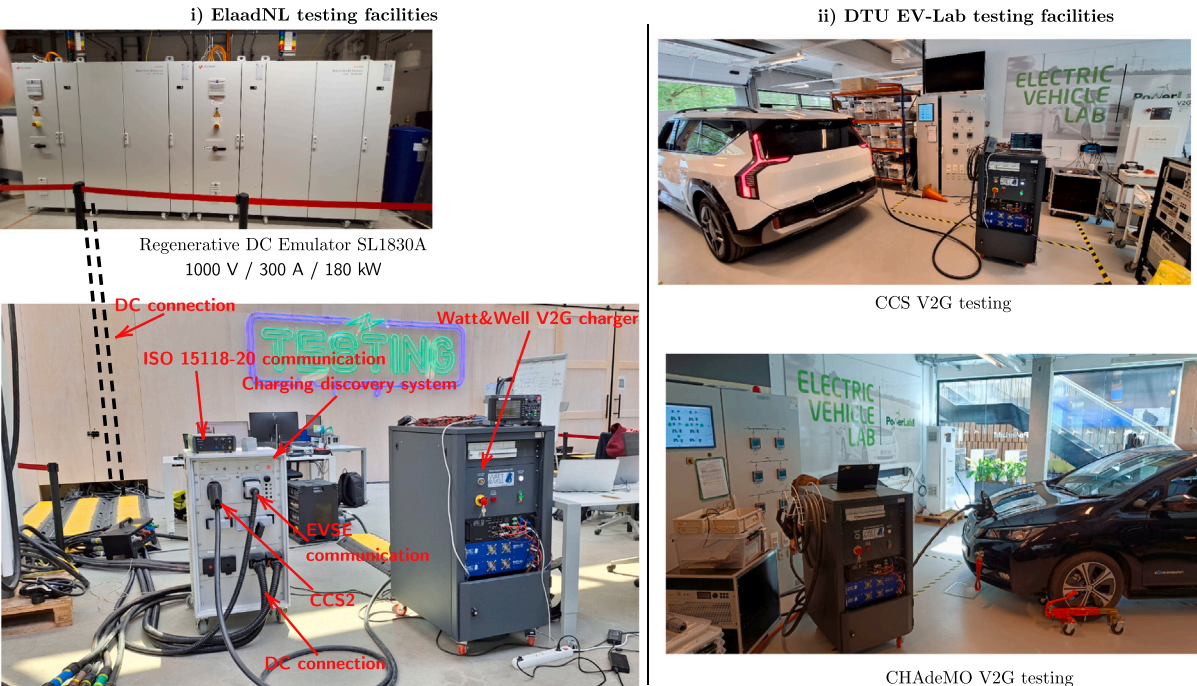
Table 1
Overview of tested bidirectional charger efficiencies. All tested chargers are 400V architecture.

Study	Year	Type	Efficiency	Physical test	Full range	Voltage/SoC
[24]	2017	CHAdeMO	64–88%	Yes	No	Yes
[18,31]	2019	CHAdeMO	0–90%	Yes	Yes	Yes
[23,30]	2020	CHAdeMO	86–88%	Yes	No	Yes
[32,33]	2024	CHAdeMO	50–90%	Yes	No	Yes
This study	2024	CHAdeMO & CCS2	0–92.8/96.9%	Yes	Yes	Yes

* This study provides data for both 400 and 800V architecture.



(a) Visualization of the methodology used for the testing campaign.



(b) Test setups for the bidirectional charging (i) ElaadNL charging discovery system acting as vehicle emulator. (ii) DTU EV-Lab test facilities with EVs.

Fig. 1. Methodology and test setups for the bidirectional charging experiments.

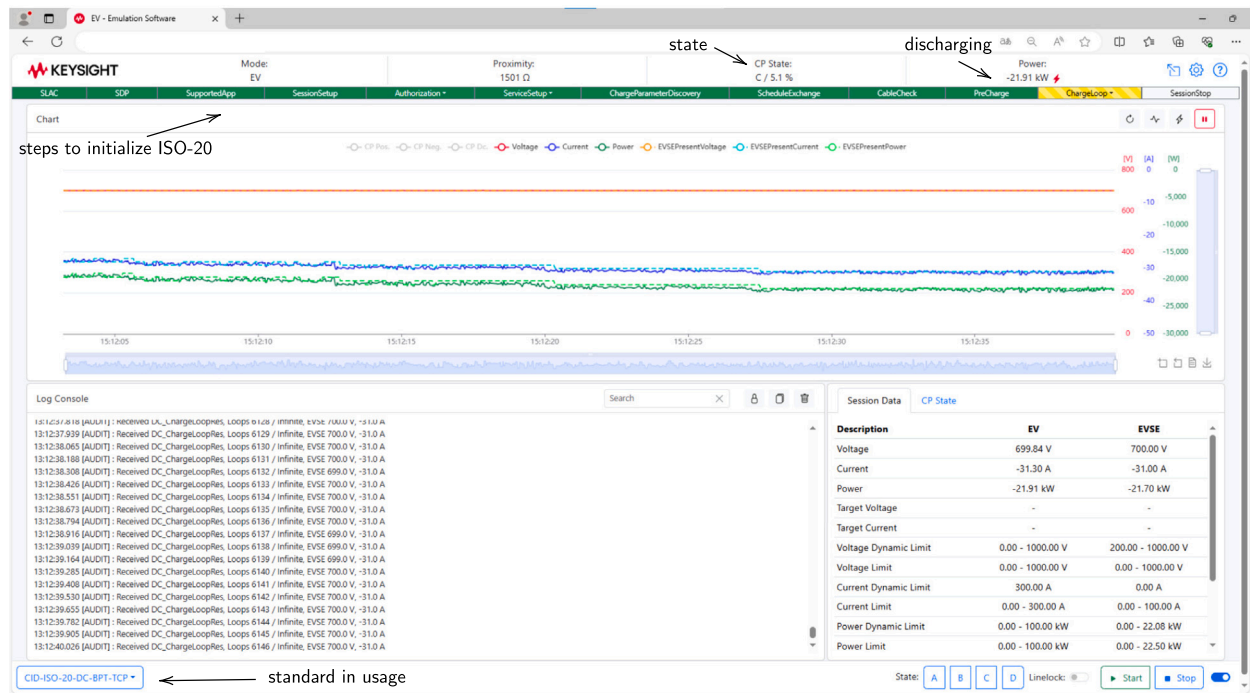


Fig. 2. Screenshot of the Keysight application interface from the charging discovery system. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Results from the testing campaign in relation to the standard used. In **bold** the successful bidirectional power transfer.

Brand	Model	Trim/year	Software version	Priority/SupportedAppProtocols	BPT working
Nissan	LEAF	40 kWh 2017	Unknown	CHAdemo	Yes
Opel	Corsa-e	Elegance 2021	psag3_ar_19.1	1 urn:din:70121:2012:MsgDef DIN 2 urn:iso:15118:2:2013:MsgDef ISO2	No
Polestar	2 LRSM	2WD 2023	P2.14.3	1 urn:din:70121:2012:MsgDef DIN 2 urn:iso:15118:2:2013:MsgDef ISO2	No
Volkswagen	ID.4 Pro	77 kWh 2023	3.5.0	1 urn:din:70121:2012:MsgDef DIN 2 urn:iso:15118:2:2013:MsgDef ISO2	No
Kia	EV9	GT line 2024	MV1.EUR.116	1 urn:din:70121:2012:MsgDef DIN 2 urn:iso:15118:2:2013:MsgDef ISO2	No
Tesla	Model Y	RWD 2023	v12 (2024.20.1)	1 urn:tesla:din:2018:MsgDef 2 urn:din:70121:2012:MsgDef DIN	No
Kia	EV9	2024	Prototype	1 urn:din:70121:2012:MsgDef DIN 2 urn:iso:15118:2:2013:MsgDef ISO2 3 urn:iso:std:iso:15118:-20:DC ISO20_DC 4 urn:iso:std:iso:15118:-20:AC_{ []redacted{} }	Yes
Kia	EV9	2024	Production	1 urn:din:70121:2012:MsgDef DIN 2 urn:iso:15118:2:2013:MsgDef ISO2	No
Keysight	CDS	N/A	N/A	urn:iso:std:iso:15118:-20:DC ISO20_DC	No - no handshake
Keysight	CCIT (Verisco)	N/A	N/A	urn:iso:std:iso:15118:-20:DC ISO20_DC	Yes - via CDS
Trialog	ComboCS4M	N/A	N/A	urn:iso:std:iso:15118:-20:DC ISO20_DC	Yes - unpowered

Investigation campaign results

ISO 15118 interoperability results

As explained previously, the implementation of ISO 15118-20 has been challenging for automakers and charger manufacturers. Thus, this result section provides an overview of the successful or not, implementation of the ISO 15118-20 standard. The testing campaign offers one-of-a-kind results from EVs to emulators. Table 2 provides information on the testing campaign carried out in Denmark and the Netherlands.

From the subjects tested, it is observed that for EVs the preferred protocol is DIN SPEC 70121, with the previous generation ISO 15118-2 not supporting bidirectional power flow, also supported. However, a commercial vehicle model (Kia EV9) was able to perform bidirectional power flow. The difference in software integration can be observed when comparing the software version of such EV model with an available production unit. Kia EV9 has ISO 15118-20 implementation for DC and AC bidirectional power flow. In addition, such an implementation was successful in achieving DC bidirectional power flow with a third-party charger and testing team, outside of the closed automaker ecosystem. A similar successful result in interoperability through ISO 15118-20 was achieved even with emulators such as Keysight CDS with the Verisco SL1550 A EV – EVSE Charging Communication Interface

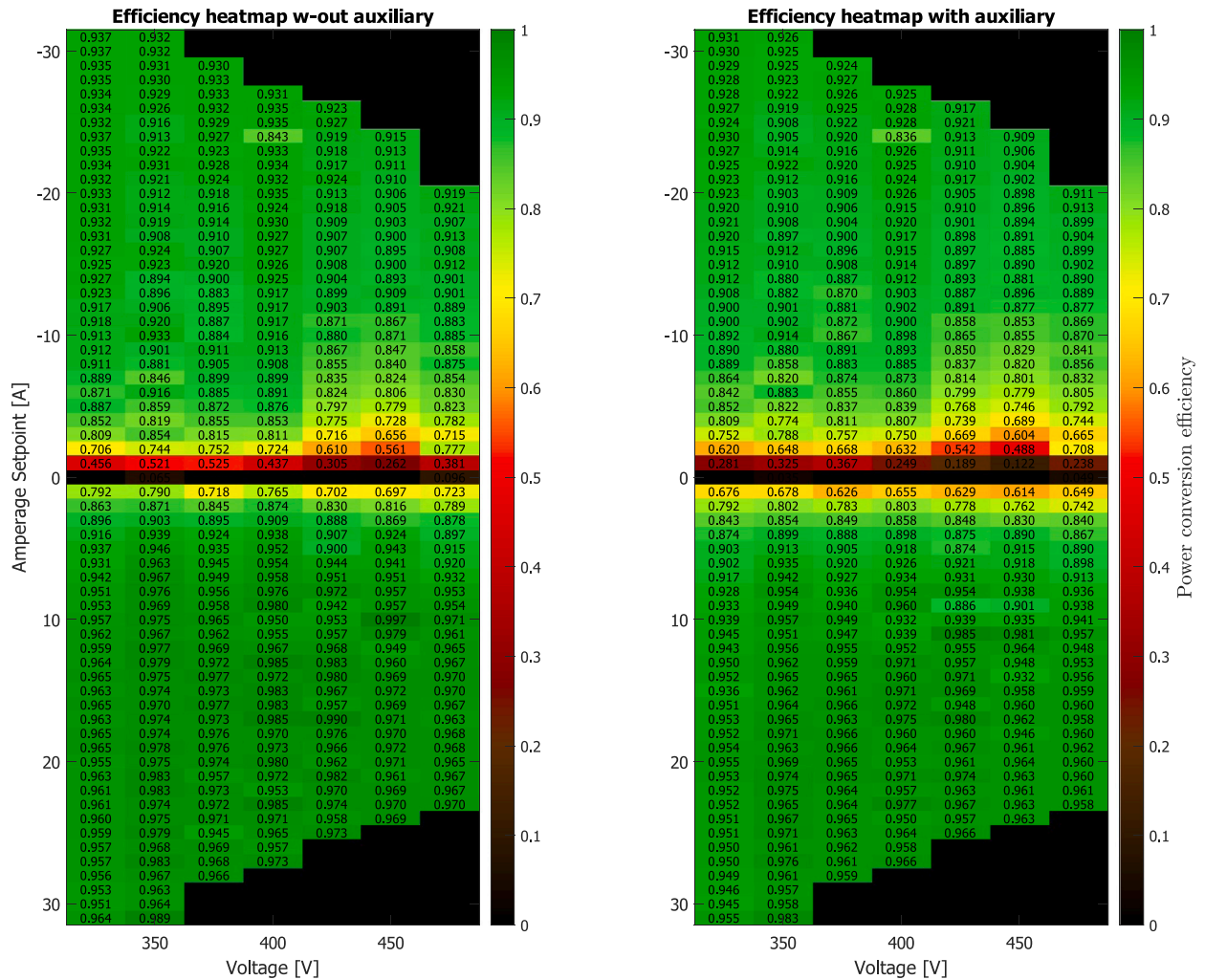


Fig. 3. Efficiency of one bidirectional power module. The voltage operation is from 325 to 475 V. A positive current indicates EV charging.

Tester (CCIT) and Trialog ComboCS4M. Consequently, such results prove that the implementation of ISO 15118-20 and the interoperability challenges are overcome. However, it should be noted that the fully compliant implementation with TLS 1.3 encryption was not successful for bidirectional power transfer. Fig. 2 provides a clear picture of the successful integration of ISO 15118-20 of the Watt&Well bidirectional charger and Keysight CDS acting as a EV emulator. The bottom left corner shows the standard in use, which is CID-ISO-20-DC-BPT-TCP. The top bar indicates the stages for ISO 15118-20 with green representing successful stages, yellow representing active stages, and gray representing noninitiated states. The steps are SLAC, SDP, SupportedApp, SessionSetup, Authorization, ServiceSetup, ChangeParameterDiscovery, ScheduleExchange, CableCheck, PreCharge, and ChargeLoop. In addition, the top screen displays the Mode: EV, Proximity pilot resistance: 1501 Ohm, the control pilot state: C/5.1% and power: -21.91 kW (discharging). The low-level control pilot state (based on IEC 61851-1) has A, B, and C options. A, B, and C correspond to not connected, connected/waiting for transaction, and charging states, respectively. The 5% duty cycle indicate that digital communication is required. The bottom right corner shows the session data between EV and EVSE. One can observe the difference in DC voltage and current between EV and EVSE. For example, the EV data show 699.84 V, -31.3 A, -21.91 kW, and the EVSE data are 700 V, -31.0 A, and -21.7 kW. The DC voltage difference ensures a power flow from EV to EVSE.

Furthermore, to complement the results of the emulator, Table 3 filters the possible options for the state-of-the-art EV/EVSE emulators

and communication sniffing products. This information is of paramount importance for technology providers to advance bidirectional capabilities. Based on publicly available data, only two manufacturers offer TLS encrypted ISO 15118-20 emulators. However, from the testing campaign with Keysight setup, it was not possible to successfully integrate TLS encryption, which is necessary for full compliance with 15118-20. To conduct a TLS 1.3 full compliance integration, the charger and vehicle should exchange certificates and decrypt the transaction message. That would be possible if the automaker shares their certificates and software houses such as Trialog update the charger software to be able to understand the communication with the vehicle. Hence, further testing of such technologies is required. Another notable observation is that while using the emulator as a man-in-the-middle between the Watt&Well bidirectional charger and the EV, it was not possible to achieve bidirectional power flow.

Bidirectional charger efficiency

The results of the bidirectional power flow provided in this section are from both the emulator acting as a vehicle and the vehicle itself. Since the connection is DC-based, the conversion efficiency is the feature of the charger where the conversion from AC to DC happens. In addition, the results reflect the average values over four measurements. As mentioned in [35], the dominant EV architecture is 400 V, however, some new models also employ 800 V. Therefore, to cover both types the bidirectional charger was used with one (400 V) and two (800 V) power

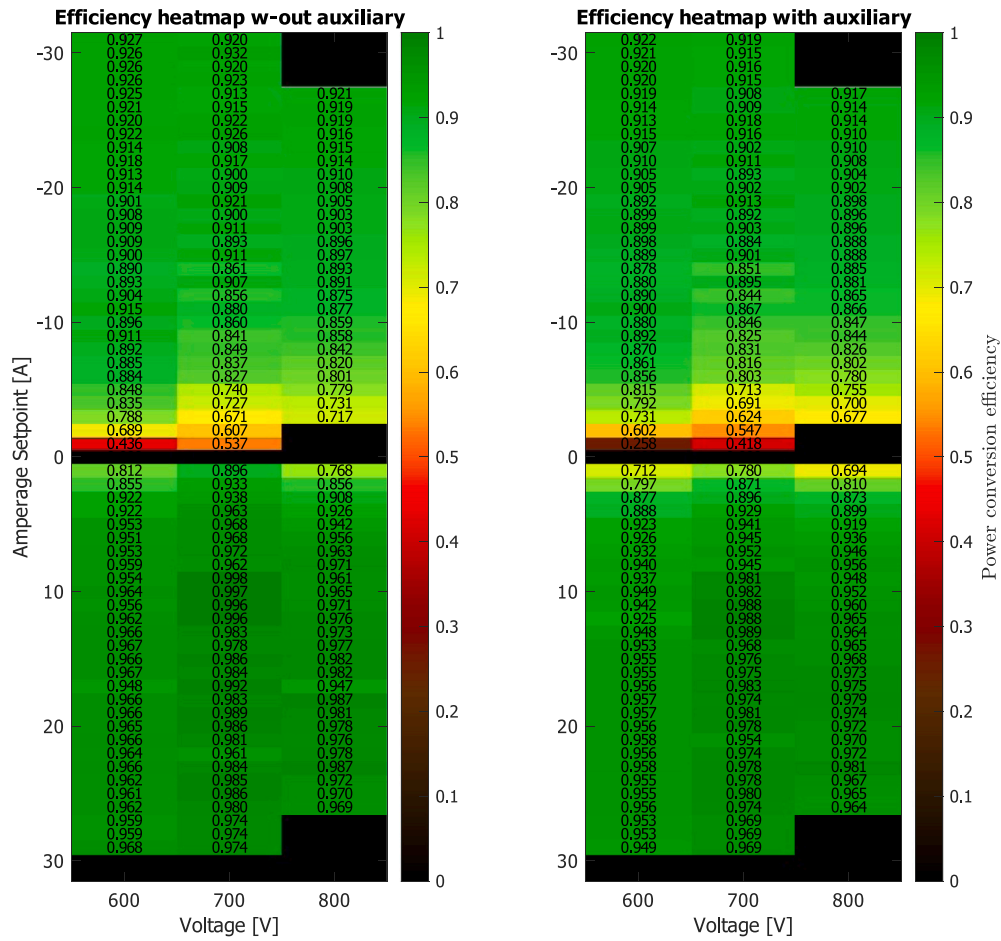


Fig. 4. Efficiency of two bidirectional power modules in series. The voltage operation is from 600–800 V. A positive current indicates EV charging.

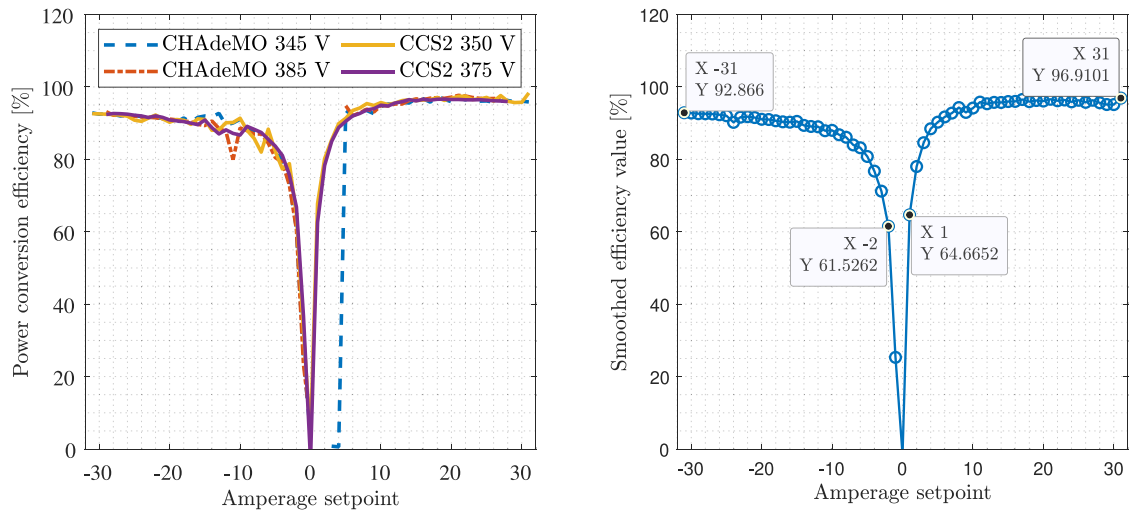


Fig. 5. (left) Efficiency of the bidirectional power module with CHAdeMO and CCS2. (right) Recommendation on the values to be used for the efficiency of bidirectional charger.

modules in series. The different voltage levels reflect the EV battery state-of-charge (SOC).

Fig. 3 displays the efficiency results for one bidirectional power module of the Watt&Well charger. The auxiliary supply of the charger is measured separately, as illustrated in Fig. 1(a). This allows for including auxiliary consumption in the efficiency considerations, which is shown in the second graph in Fig. 3. It is important to note that

the efficiency of discharging (0 to −31 A) is lower than the efficiency of charging (0 to 31 A). The efficiency values follow a well-known parabolic curve, but deviate from previous studies as they show better efficiency at low currents. That is essential for the feasibility of the business model, as most of the time the bidirectional charger would work in low current charging / discharge mode [36]. Operating at low currents, specifically at discharging, yields significantly low efficiency values;

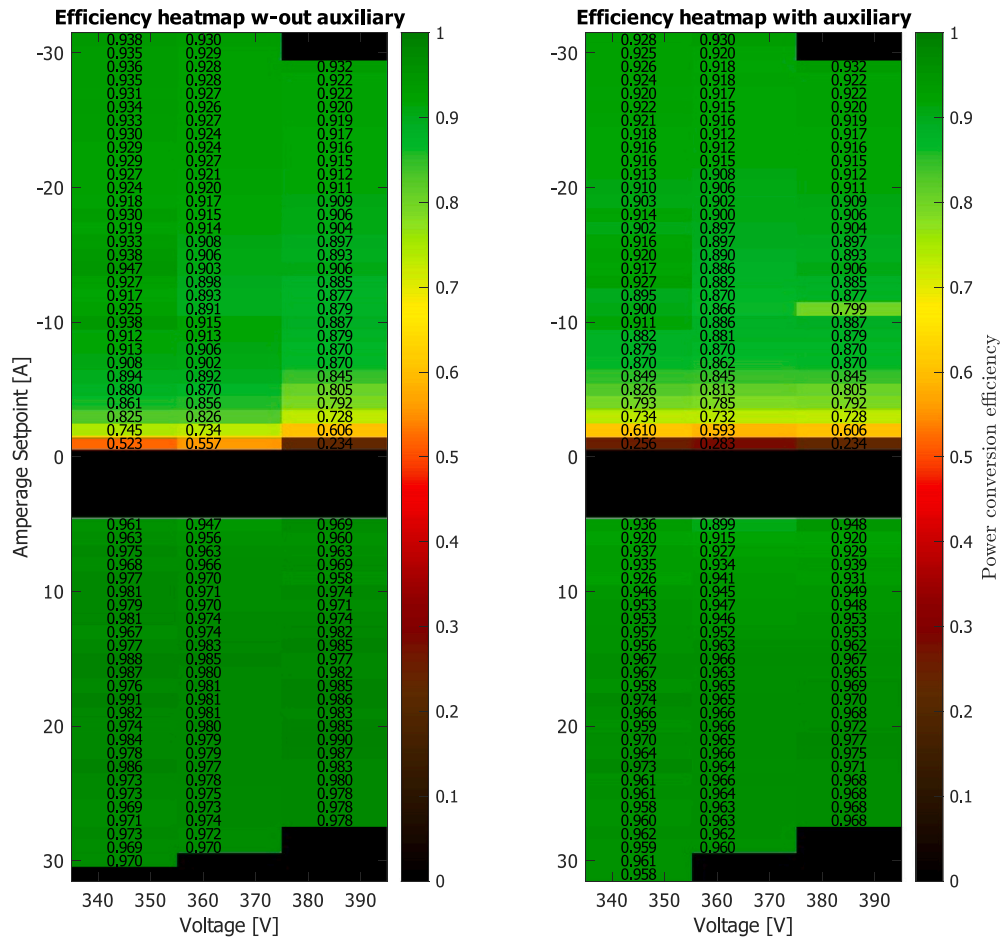


Fig. 6. Efficiency of the bidirectional power module with CHAdeMO protocol using a Nissan LEAF e+. A positive current indicates EV charging.

Table 3

Overview of EV/EVSE emulators and communication sniffing products. In this table, *All* in the protocols column refers to support for DIN SPEC 70121, ISO 15118-2 and ISO 15118-20. *TLS decrypt* refers to whether the system in question claims to be able to decrypt TLS-encrypted communication.

No.	Brand	Name	Model no.	Protocols	Features	Powered testing support	TLS decrypt
1	Trialog	ComboCS4M		All	EV emulation	No native power support. Third-party integration possible	N/A
2	Trialog	Sniffer		All	Sniffing	N/A	No
3	Keysight	EV-EVSE charging communication interface tester	SL1550A(HW) SL1470A (SW)	All	EV emulation EVSE emulation	Native support for keysight charging discovery system	N/A
4	Keysight	CCS Charging protocol tracer	SL1556A (HW) SL1487A (SW)	All	Sniffing	N/A	Yes
5	Keysight	Charging discovery system	SL1047A (HW) SL1093A (SW)	All	EV emulation EVSE emulation MitM	AC emulation: SL1200A DC emulation: SL1800A	No
6	Comemso	EVCA flex		All	EV emulation EVSE emulation MitM	No native power support. Third-party integration possible	Presumed
7	Comemso	EVCA multi mobile		All	EV emulation EVSE emulation MitM	No native power support. Third-party integration possible	Yes
8	Comemso	EVCA ComOnly		All	EV emulation EVSE emulation Sniffing	No	Presumed
9	IoTecha	PilotShark		All	Sniffing	N/A	Claimed
10	VertexCom	GreenPHY sniffer	MST216D-D01D	No ISO-20	Sniffing	N/A	No
11	Applus	CCS V2G decoder		No ISO-20	Sniffing	N/A	No

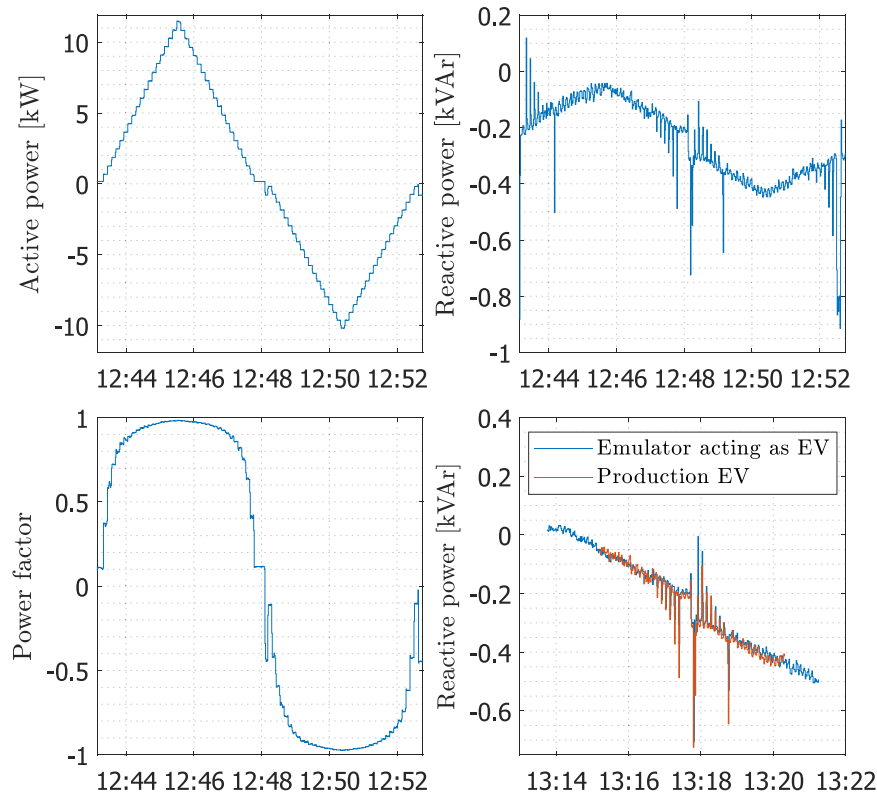


Fig. 7. Comparison of bidirectional charger power quality for 575 V* when in operation with the emulator and the production vehicle. * 575 V corresponds to 50% SOC for the vehicle. At the bottom right corner, the emulator power stretches between -22 to 22 kW, while the vehicle from -11 to 11 kW.

however, increasing the current (charging or discharging) certainly improves the efficiency.

Fig. 4 extends the test towards the 800 V EV architecture. The efficiency appears to be similar to the previous 400 V tests. The discharge process is less efficient than the charging process. Moreover, the Watt&Well bidirectional charger also offers the option to perform bidirectional power flow with the CHAdeMO protocol. Thus, Fig. 5 provides a detailed comparison of the power conversion efficiency between CHAdeMO and CCS2. As expected, the efficiency is similar because it reflects the characteristics of power electronics, not the standard (or protocol).

However, there is a difference in the available charging current of the CHAdeMO protocol. The Nisan Leaf mandates a minimum charging current of 5 A through the CHAdeMO communication [37], any charging attempts below this threshold is thereby interrupted by the EVSE. The results show that the efficiency is better than previously tested bidirectional chargers. One can note that for CHAdeMO the efficiency is around 60 to 93% for discharging and 90 to 97% the charging efficiency. A full picture can be found in Fig. 6. Furthermore, based on Fig. 5(b), we provide the mathematical model for the generic efficiency curve and the fitted parameters. We model losses as $P_{\text{loss}}(x) = P_0 + bx^2$, giving $\eta(x) = x/(x + k_0 + k_2x^2)$ with $x = |I|$. Parameters are obtained by ordinary least squares (OLS) on $y = x(1/\eta - 1) = k_0 + k_2x^2$ using all $|I| > 0$ data.

- Pooled fit: $k_0 = 0.9527$ [0.7356, 1.1699], $k_2 = 9.997 \times 10^{-4}$ [5.139×10^{-4} , 1.485×10^{-3}], $R^2 = 0.775$, RMSE = 0.0518.
- Direction-specific fits (recommended for simulations): charge $k_0 = 0.4817$ [0.4199, 0.5435], $k_2 = 8.741 \times 10^{-4}$ [7.359×10^{-4} , 1.012×10^{-3}], $R^2 = 0.982$, RMSE = 0.00871;
- Direction-specific fits (recommended for simulations): discharge $k_0 = 1.4238$ [1.250, 1.598], $k_2 = 1.125 \times 10^{-3}$ [7.364×10^{-4} , 1.514×10^{-3}], $R^2 = 0.939$, RMSE = 0.0321.

Table A.1

ISO 15118-20 power transfer modes adapted from ISO 15118-20 table 204.

ServiceID	ServiceName	Description
1	AC	AC energy transfer, physical layer according to ISO 15118-3 and ISO 15118-8
2	DC	DC energy transfer, physical layer according to ISO 15118-3 and ISO 15118-8
3	WPT	Wireless power transfer, physical layer according to ISO 15118-8
4	DC_ACDP	DC Charging with ACDP, physical layer according to ISO 15118-8
5	AC_BPT	AC Charging with BPT, physical layer according to ISO 15118-3 and ISO 15118-8
6	DC_BPT	DC Charging with BPT, physical layer according to ISO 15118-3 and ISO 15118-8
7	DC_ACDP_BPT	DC Charging with ACDP and BPT, physical layer according to ISO 15118-8

Finally, Fig. 7 provides a comprehensive view on the power quality measurements such as reactive power and power factor on the entire range. The measurements are consistent with such values. The reactive power consumption during charging process is better when compared to commercial EV onboard chargers. Power factor (PF) is the ratio of active power to apparent power. A PF of 1 means that all the power supplied is being used to do useful work, while a PF of less than 1 indicates that some of the power is being wasted. Consequently, a lower PF can be caused by inductive loads (e.g. electric motors), reactive power, and nonlinear loads (e.g. electronic equipment). The power factor is above 0.9 after 3 kW. The low-voltage grid code demands to be 0.9 inductive as mandated in EU Commission Regulation 2016/1388 for the connection of demand to the low-voltage grid [38].

Table A.2

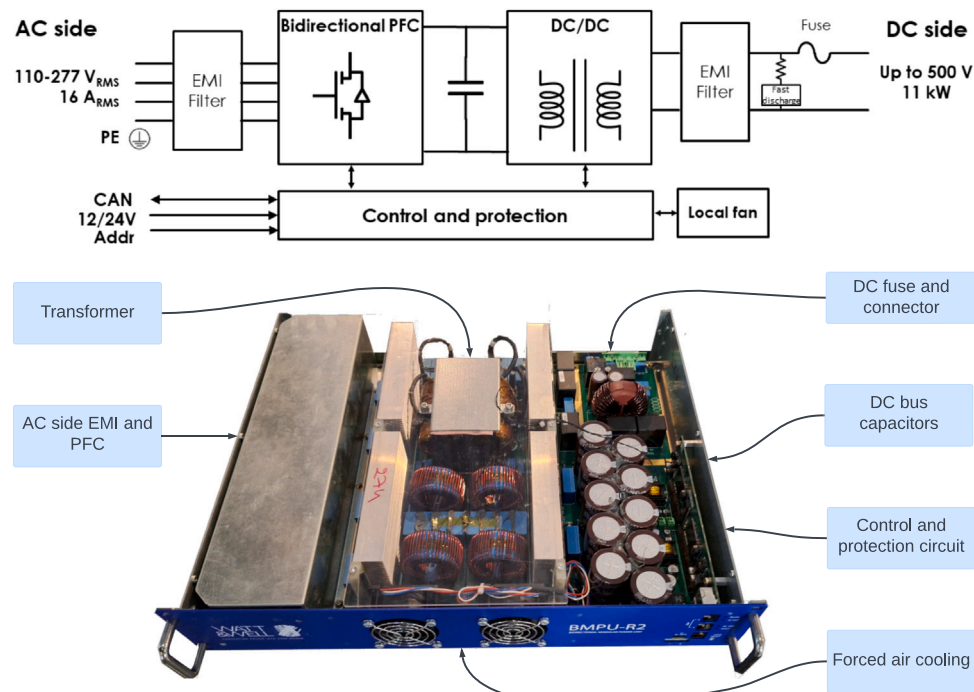
Overview of measurement equipment's, the selected measurement and range and the applicable uncertainty for that range.

Brand	Model	Measurement	Uncertainty
Chauvin Arnux	PEL 104/106	AC voltage AC current (MA193) Active power Reactive power Power factor	$\pm 0.2\%$ R ± 0.2 V $\pm 0.2\%$ R $\pm 0.02\%$ Inom $\pm 0.5\%$ R $\pm 0.005\%$ Pnom $\pm 1\%$ R $\pm 0.01\%$ Qnom ± 0.05
Chauvin Arnux	L452	Voltage ± 1 V	1 mV $\pm(0.5\% + 1$ cts)
Hioki	LR8431-20	Voltage 1 V	$\pm 0.1\%$
Hioki	P9000-01	Voltage 1:1000	$\pm 0.5\%$
Keysight	1146B	Current 10 mV/A	4%

Table A.3

Watt&Well charger limits for various configurations* in charging and discharge mode [39]. *Voltage and current refer to the DC side.

	Charging			Discharging		
	Current [A]	Voltage [V]	Power [kW]	Current [A]	Voltage [V]	Power [kW]
Individual BMPU	0 to 30	150 to 500	0 to 10.5	-32 to 0	250 to 500	-11 to 0
Parallel BMPU	0 to 60	150 to 500	0 to 21	-64 to 0	250 to 500	-22 to 0
Series BMPU	0 to 30	300 to 1000	0 to 21	-32 to 0	500 to 1000	-22 to 0

**Fig. A.1.** Block diagram and internal overview of the Watt&Well BMPU.

Conclusions

This study provides a comprehensive experimental validation of bidirectional EV charging using the ISO 15118-20 standard with both CCS2 and CHAdeMO protocols. We successfully demonstrated the interoperability of the Watt&Well bidirectional charger with emulators and a commercial vehicle (Kia EV9), overcoming several longstanding challenges in ISO 15118-20 implementation. Our findings mark a critical step towards achieving a more interoperable and efficient V2G ecosystem. These results emphasize that we can achieve ISO 15118-20 interoperability as long as each actor builds their system according to the standard. However, the full TLS compliance was not achieved. Absence of an industry-wide mechanism to distribute/cross-sign OEM Root-CAs into EVSE trust-stores results in TLS failures despite otherwise compliant stacks.

A key contribution of this research is the in-depth analysis of the charging and discharging efficiencies of the bidirectional charger. The results highlight that while the charger achieves a high charging efficiency (up to 97% for both CCS2 and the CHAdeMO protocols), the discharging efficiency is significantly lower, ranging from 60% to 93%. This efficiency disparity, particularly at lower power levels, emphasizes the need for optimization in discharging modes, especially as discharging is a critical feature for V2G applications. The parabolic efficiency curve observed shows that low current discharges (around 1–2 A) perform poorly, but efficiency increases significantly at higher current levels. To facilitate and reduce computational capabilities, we have proposed a generic efficiency curve that can be used by scholars or professionals in their calculations. Subsequently, it is necessary to highlight that the data showcased in this paper are biased towards the

power electronics designed and built by Watt&Well for their commercial application. However, the efficiency maps reported are expected to transfer to other modern off-board V2G chargers for three reasons: (i) the underlying two-stage AC front-end + isolated DC-DC topology is industry-standard (Fig. 4); (ii) protocol choice (CHAdeMO vs. CCS2/ISO 15118-20) did not materially affect conversion efficiency at matched DC setpoints; and (iii) testing covered both 400 V and 800 V architectures across the full operating area, with auxiliary consumption included.

Interestingly, the results demonstrate that efficiency has improved, in comparison to previous CHAdeMO generation, in low-current charging and discharging modes, which is especially beneficial for real-world applications. This makes the technology more attractive for smart charging and V2G applications depending on the use case. For example, a behind-the-meter V2G use case would require low power operation, while an in front-of-the-meter V2G use case would require high-power operation. Further comparison between CHAdeMO and CCS2 showed that the protocol itself did not significantly affect efficiency, as both systems performed similarly due to the underlying power electronics. However, the CHAdeMO protocol exhibited a limitation in consuming energy at very low currents (0–4 A), which could be a consideration in optimizing charger operation. The power quality measurements, including reactive power and power factor, further supported the viability of this bidirectional charger for integration into the grid. The power factor remained above 0.9 after 3 kW, meeting grid code requirements and ensuring efficient energy transfer without causing undue stress on the power system.

These findings contribute valuable insights into the efficiency limitations and strengths of current bidirectional charging technologies. Future research should focus on enhancing the discharging efficiency and further enhancing the interoperability of chargers and EVs, particularly with TLS encryption, to make the technology more commercially viable and grid-friendly. In addition, the investigation will focus on the delivery of grid services with bidirectional chargers.

CRediT authorship contribution statement

Kristian Sevdari: Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kristoffer Laust Pedersen:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Gabriel Fabbri:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Data curation. **Rasmus Meier Knudsen:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis. **Abdesslam Alami:** Validation, Supervision, Resources, Project administration, Investigation. **Jan Engelhardt:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis. **Mattia Marinelli:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The work in this paper is supported by the research projects EV4EU (Horizon Europe grant no. 101056765) and FLOW (Horizon Europe grant no. 101056730). The authors thank Jan Karens and Robert Italiaander from ElaadNL, Redha Meghnous, and Nadim Sakr from Watt&Well for helping on such a testing campaign.

Appendix. Metadata

See Tables A.1–A.3 and Fig. A.1.

Data availability

Data will be made available on request.

References

- [1] Sovacool BK, Axsen J, Kempton W. The future promise of vehicle-to-grid (V2G) integration: a sociotechnical review and research agenda. *Annu Rev Environ Resour* 2017;42(1):377–406.
- [2] Striani S, Sevdari K, Calearo L, Andersen PB, Marinelli M. Barriers and solutions for evs integration in the distribution grid. In: 2021 56th international universities power engineering conference. UPEC, IEEE; 2021, p. 1–6.
- [3] Englberger S, Gamra KA, Tepe B, Schreiber M, Jossen A, Hesse H. Electric vehicle multi-use: Optimizing multiple value streams using mobile storage systems in a vehicle-to-grid context. *Appl Energy* 2021;304:117862.
- [4] Ziras C, Thingvad M, Fog T, Yousefi G, Weckesser T. An empirical analysis of electric vehicle charging behavior based on real danish residential charging data. *Electr Power Syst Res* 2024;234:110556.
- [5] Bollerslev J, Andersen PB, Jensen TV, Marinelli M, Thingvad A, Calearo L, Weckesser T. Coincidence factors for domestic EV charging from driving and plug-in behavior. *IEEE Trans Transp Electrification* 2021;8(1):808–19.
- [6] Klobasa M, Kühnbach M. Study on the "Potential of a full EV-power-system-integration in Europe & how to realise it" Imprint Potential of a full EV-power-system-integration in Europe and how to realise it Project coordination Responsible for content. tech. rep.
- [7] Secchi M, Macii D, Barchi G, Marinelli M. Centralised vehicle-to-grid smart charging supported by PV generation for power variance minimisation at the transformer: A user's perspective analysis. *ETransportation* 2025;100394.
- [8] Lund H, Kempton W. Integration of renewable energy into the transport and electricity sectors through V2G. *Energy Policy* 2008;36(9):3578–87.
- [9] Marinelli M, Calearo L, Ried S, Pfab X, Cabrera JCD, Spalthoff C, Braun M, Sæle H, Torsæter BN, Astero P, et al. Electric vehicles demonstration projects-an overview across europe. In: 2020 55th international universities power engineering conference. UPEC, IEEE; 2020, p. 1–6.
- [10] Calearo L, Marinelli M. Profitability of frequency regulation by electric vehicles in Denmark and Japan considering battery degradation costs. *World Electr Veh J* 2020;11(3):48.
- [11] Brey BD. Smart solar charging: bi-directional AC charging (V2G) in the Netherlands. *J Energy Power Eng* 2017-07-28;11(7).
- [12] Thingvad A, Calearo L, Andersen PB, Marinelli M, Neimeh M, Suzuki K, Murai K. Value of V2G frequency regulation in great britain considering real driving data. In: 2019 IEEE PES innovative smart grid technologies europe (ISGT-europe). IEEE; 2019, p. 1–5.
- [13] Kempton W, Tomic J, Letendre S, Brooks A, Lipman T. Vehicle-to-grid power: battery, hybrid, and fuel cell vehicles as resources for distributed electric power in california. 2001.
- [14] Thingvad A, Calearo L, Andersen PB, Marinelli M. Empirical capacity measurements of electric vehicles subject to battery degradation from V2G services. *IEEE Trans Veh Technol* 2021;70(8):7547–57.
- [15] Gong J, Wasylowski D, Figgner J, Bihn S, Rücker F, Ringbeck F, Sauer DU. Quantifying the impact of V2X operation on electric vehicle battery degradation: An experimental evaluation. *ETransportation* 2024;20:100316.
- [16] Huber J, Schaul E, Jung D, Weinhardt C. Quo vadis smart charging? A literature review and expert survey on technical potentials and user acceptance of smart charging systems. *World Electr Veh J* 2019;10(4):85.
- [17] Pearre NS, Ribberink H. Review of research on V2X technologies, strategies, and operations. *Renew Sustain Energy Rev* 2019;105:61–70.
- [18] Thingvad A, Ziras C, Marinelli M. Economic value of electric vehicle reserve provision in the nordic countries under driving requirements and charger losses. *J Energy Storage* 2019;21:826–34.
- [19] Sevdari K, Calearo L, Andersen PB, Marinelli M. Ancillary services and electric vehicles: An overview from charging clusters and chargers technology perspectives. *Renew Sustain Energy Rev* 2022;167:112666.
- [20] Secretary IC. Road vehicles — Vehicle to grid communication interface - Part 20: 2nd generation network layer and application layer requirements. Standard ISO 15118-20:2022, Geneva, CH: International Organization for Standardization; 2022.
- [21] Mültin M. The new features and timeline for ISO 15118-20. 2020, URL <https://www.switch-ev.com/blog/new-features-and-timeline-for-iso15118-20>.
- [22] Sakr N, Iglesias R. Universal and versatile charger for electric vehicle battery. US Patent App. 17/676,991, Google Patents; 2022.

- [23] Schram WL, AlSkaif T, Lampropoulos I, Henein S, Van Sark WG. On the trade-off between environmental and economic objectives in community energy storage operational optimization. *IEEE Trans Sustain Energy* 2020;11(4):2653–61.
- [24] Apostolaki-Iosifidou E, Codani P, Kempton W. Measurement of power loss during electric vehicle charging and discharging. *Energy* 2017;127:730–42.
- [25] Wouters H, Martinez W. Bidirectional on-board chargers for electric vehicles: State-of-the-art and future trends. *IEEE Trans Power Electron* 2023.
- [26] Khaligh A, Dantonio M. Global Trends in High-Power On-Board Chargers for Electric Vehicles. *IEEE Trans Veh Technol* 2019;68(4):3306–24.
- [27] Rivera S, Kouro S, Vazquez S, Goetz SM, Lizana R, Romero-Cadaval E. Electric Vehicle Charging Infrastructure: From Grid to Battery. *IEEE Ind Electron Mag* 2021;15(2):37–51.
- [28] Sevdari K, Calearo L, Bakken BH, Andersen PB, Marinelli M. Experimental validation of onboard electric vehicle chargers to improve the efficiency of smart charging operation. *Sustain Energy Technol Assessments* 2023;60:103512.
- [29] Yasko M, Driesen J, Martinez W. Efficiency measurement and maximization for EV charging technologies. In: 2024 IEEE transportation electrification conference and expo, ITEC 2024. IEEE; 2024, p. 1–5.
- [30] Schram W, Brinkel N, Smink G, Van Wijk T, Van Sark W. Empirical evaluation of V2G round-trip efficiency. In: 2020 international conference on smart energy systems and technologies. SEST, IEEE; 2020, p. 1–6.
- [31] Zecchino A, Thingvad A, Andersen PB, Marinelli M. Test and modelling of commercial V2G chademo chargers to assess the suitability for grid services. *World Electr Veh J* 2019;10(2):21.
- [32] Vanderwegen W, Martinez W. Enhancing EV Charging Efficiency: Conversion of bidirectional two-stage on-board chargers to single-stage topologies. In: 2024 IEEE 10th international power electronics and motion control conference, IPEMC 2024 ECCE Asia. IEEE; 2024, p. 4146–9.
- [33] Iwafune Y, Kawai T. Data analysis and estimation of the conversion efficiency of bidirectional EV chargers using home energy management systems data. *Smart Energy* 2024;100145.
- [34] Senol M, Bayram IS, Campos-Gaona D, Sevdari K, Gehrke O, Pepper B, Galloway S. Measurement-based harmonic analysis of electric vehicle smart charging. In: 2024 IEEE transportation electrification conference & expo. IEEE; 2024.
- [35] Sevdari K, Marinelli M, Pastorelli F. Overview of EV battery types and degradation measurement for renault zoe NMC batteries. In: Proceedings of 2024 international conference on renewable energies and smart technologies. REST, IEEE; 2024.
- [36] Tveit MH, Sevdari K, Marinelli M, Calearo L. Behind-the-meter residential electric vehicle smart charging strategies: Danish cases. In: 2022 international conference on renewable energies and smart technologies. REST, 1, IEEE; 2022, p. 1–5.
- [37] ICE. IEC IEC 62196-2:2022 - Plugs, socket-outlets, vehicle connectors and vehicle inlets - Conductive charging of electric vehicles - Part 2: Dimensional compatibility requirements for AC pin and contact-tube accessories. Standard IEC 62196-2:2022, International Electrotechnical Commission; 2022.
- [38] Denmark GP. Guide for grid connection of demand installations to the low-voltage grid. *Tech. Rep.*, 2022, p. 1–42.
- [39] Watt&Well. EVI -Technical Reference Manual rev AE. 2024.